

PHYSICS

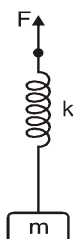
TARGET : JEE- Advanced

CAPS-5

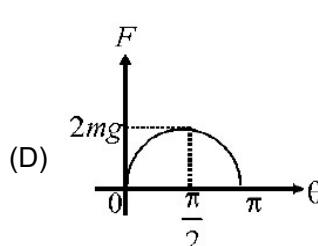
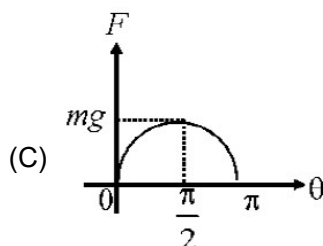
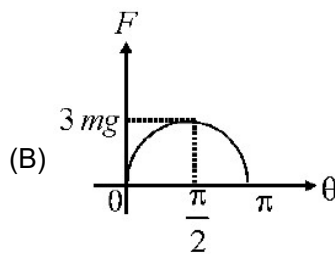
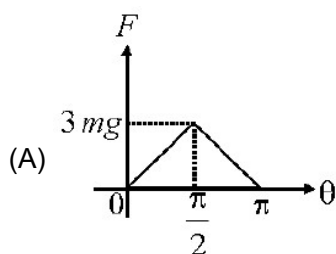
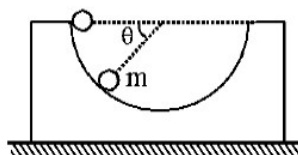
WORK POWER ENERGY

SCQ (Single Correct Type) :

1. A block of mass m is connected to a spring (spring constant k). Initially the block is at rest and the spring is in its natural length. Now the system is released in gravitational field and a variable force F is applied on the upper end of the spring such that the downward acceleration of the block is given as $a = g - \alpha t$, where t is time elapses and $\alpha = 1 \text{ m/s}^3$, the velocity of the point of application of the force is :



- (A) $\frac{m\alpha}{k} - gt + \frac{\alpha t^2}{2}$ (B) $\frac{m\alpha}{k} + gt + \frac{\alpha t^2}{2}$ (C) $\frac{m\alpha}{k} - gt - \frac{\alpha t^2}{2}$ (D) $\frac{m\alpha}{k} - gt - \alpha t^2$
2. In the shown fig., small mass m starts sliding down a smooth and stationary circular track. Which of the following graph best represents the variation of magnitude of the force applied by the track on the mass and the angle θ :

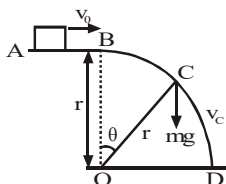


3. A body starts from rest and acquires a velocity V in time T . The work done by constant force on the body in time t is proportional to
- (A) $\frac{V}{T}t$ (B) $\frac{V^2t^2}{T}$ (C) $\frac{V^2t}{T^2}$ (D) $\frac{V^2t^2}{T^2}$
4. A certain system has potential energy given by the function $U(x) = -ax^2 + bx^4$ with constants, $a, b > 0$. Which of the following is an unstable equilibrium point?
- (A) 0 (B) $\sqrt{a/2b}$ (C) $-\sqrt{a/2b}$ (D) $\sqrt{a/b}$
5. The potential energy of a 1 kg particle free to move along the x-axis is given by

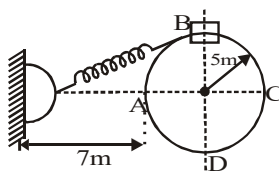
$$V(x) = \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \text{ J}$$

The total mechanical energy of the particle is 2 J. Then, the maximum speed (in m/s) is

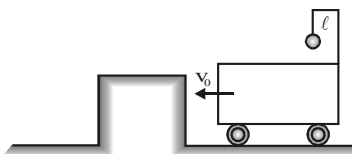
- (A) $\frac{3}{\sqrt{2}}$ (B) $3\sqrt{2}$ (C) $\frac{9}{2}$ (D) 2
6. A small block slides with velocity $0.5\sqrt{gr}$ on the horizontal frictionless surface as shown in the figure. The block leaves the surface at point C. The angle θ in the figure is :



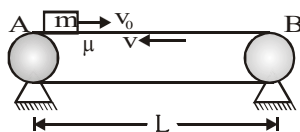
- (A) $\cos^{-1} \frac{4}{9}$ (B) $\cos^{-1} \frac{3}{4}$ (C) $\cos^{-1} \frac{1}{2}$ (D) $\cos^{-1} \frac{4}{5}$
7. A collar 'B' of mass 2 kg is constrained to move along a horizontal smooth and fixed circular track of radius 5m. The spring lying in the plane of the circular track and having spring constant 200 N/m is undeformed when the collar is at 'A'. If the collar starts from rest at B' the normal reaction exerted by the track on the collar when it passes through 'A' is :



- (A) 360 N (B) 720 N (C) 1440 N (D) 2880 N
8. A bob is suspended from a crane by a cable of length $\ell = 5$ m. The crane and load are moving at a constant speed v_0 . The crane is stopped by a bumper and the bob on the cable swings out an angle of 60° . The initial speed v_0 is— ($g = 9.8 \text{ m/s}^2$)



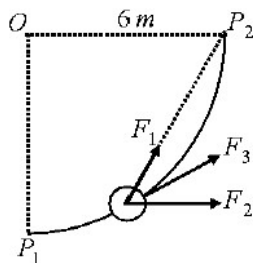
- (A) 10 m/s (B) 7 m/s (C) 4 m/s (D) 2 m/s
9. With what minimum velocity v_0 should block be projected from left end A towards end B such that it reaches the other end B of conveyer belt moving with constant velocity v ? Friction coefficient between block and belt is μ .



- (A) $\sqrt{\mu g L}$ (B) $\sqrt{2\mu g L}$ (C) $\sqrt{3\mu g L}$ (D) $2\sqrt{\mu g L}$
10. Length ℓ of a chain of mass M hangs from a pulley. If it is wound such that half of the chain remains overhung, the work done by the external agent is equal to
- (A) $\frac{Mg\ell}{2}$ (B) $\frac{3}{4}Mg\ell$ (C) $\frac{3}{8}Mg\ell$ (D) None of these
11. Kinetic energy of a particle moving in a straight line is proportional to the time t . The magnitude of the force acting on the particle is
- (A) directly proportional to the speed of the particle
 (B) inversely proportional to $\sqrt{t}$
 (C) inversely proportional to the speed of the particle
 (D) directly proportional to $\sqrt{t}$

MCQ (One or more than one correct) :

12. A smooth track in form of a quarter circle of radius 6 m lies in the vertical plane, A particle moves from P_1 to P_2 under the action of forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$. Force $\vec{F}_1$ is always towards P_2 and is always 20 N in magnitude. Force $\vec{F}_2$ is always act horizontally and is always 30 N in magnitude. Force $\vec{F}_3$ always acts tangentially to the track and is of magnitude 15 N. select the correct magnitude.



- (A) Work done by $\vec{F}_1$ is $120\sqrt{2}$ J (B) Work done by $\vec{F}_2$ is 180 J
 (C) Work done by $\vec{F}_3$ is 45π J (D) $\vec{F}_3$ is conservative in nature.

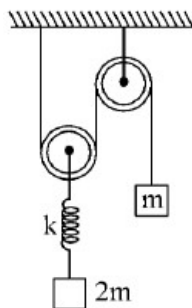
Comprehension Type Question:

The potential energy at a point, relative to the reference point is defined as the negative of work done by the conservative force as the object moves from the reference point to the point considered. The value of potential energy at the reference point itself can be set equal to zero because we are always concerned only with differences of potential energy between two points and the associated change of kinetic energy. A particle A is fixed at origin of a fixed coordinate system. Another particle B which is free to move experiences an force $\vec{F} = \left(-\frac{2\alpha}{r^3} + \frac{\beta}{r^2} \right) \hat{r}$ due to particle A where $\vec{r}$ is the position vector of particle B relative to A. It is given that the force is conservative in nature and potential energy at infinity is zero. If B has to be removed from the influence of A, energy has to be supplied for such a process. The ionization energy E_0 is work that has to be done by an external agent to move the particle from a distance r_0 to infinity slowly. Here r_0 is the equilibrium position of the particle.

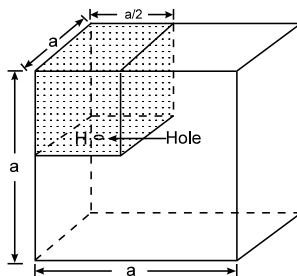
13. What is potential energy function of particle as function of r .
- (A) $\frac{\alpha}{r^2} - \frac{\beta}{r}$ (B) $-\frac{\alpha}{r^2} + \frac{\beta}{r}$ (C) $-\frac{\alpha}{r^2} - \frac{\beta}{r}$ (D) $\frac{\alpha}{r^2} + \frac{\beta}{r}$
14. Find the ionization energy E_0 of the particle B.
- (A) $\frac{\beta^2}{2\alpha}$ (B) $\frac{2\beta^2}{\alpha}$ (C) $\frac{\beta^2}{4\alpha}$ (D) $\frac{\beta^2}{\alpha}$
15. If particle B is transferred slowly from point $P_1(\sqrt{2}r_0, \sqrt{2}r_0)$ to point $P_2\left(\frac{r_0}{\sqrt{2}}, \frac{r_0}{\sqrt{2}}\right)$ in the xy-plane by an external agent, calculate work required to be done by it in the process.
- (A) $\frac{9\beta^2}{64\alpha}$ (B) $\frac{\beta^2}{16\alpha}$ (C) $\frac{\beta^2}{64\alpha}$ (D) None of these

Numerical based Questions :

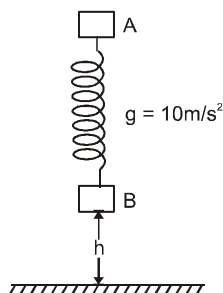
16. In the figure shown, string is light and inextensible, spring is light and ideal, the pulleys are light and frictionless. Initially the spring is in its natural length and the string just taut. Now the system is released from rest. The maximum extension in the spring during consequent motion is $\frac{nmg}{k}$ Find n



17. The figure shows a hollow cube of side 'a' of volume V. There is a small chamber of volume $\frac{V}{4}$ in the cube as shown. This chamber is completely filled by m kg of water. Water leaks through a hole H and spreads in the whole cube. Then the work done by gravity in this process assuming that the complete water finally lies at the bottom of the cube is :



18. From what minimum height 'h' in metre must the system be released when spring is in its natural length as shown in the figure. So that after perfectly inelastic collision. ($e = 0$), of block B, with ground, B may be lifted off ground. (Take $k = 40 \text{ N/m}$, $g = 10 \text{ m/s}^2$, $m_A = 2 \text{ kg}$, $m_B = 4 \text{ kg}$)



19. A pump motor is used to deliver water at a certain rate from a given pipe. To obtain "n" times water from the same pipe in the same time, the factor by which the power of the motor should be increased is:

Matrix Match Type :

20. A particle of mass 'm' moves under a conservative force with potential energy $U(x) = \frac{cx}{x^2 + a^2}$ where c & a are positive constants.

Column-I

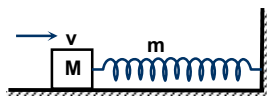
- (A) Position of unstable equilibrium is
(B) For the particle to be confined in the region the velocity of particle must be
(C) For particle to reach $x = +\infty$, velocity of particle must be
(D) For the particle to reach $x = -\infty$, velocity of particle must be

Column-II

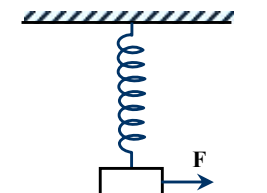
- (P) Less than $\sqrt{\frac{c}{ma}}$
(Q) More than $\sqrt{\frac{c}{ma}}$
(R) More than $\sqrt{\frac{2c}{ma}}$
(S) $x = -a$
(T) $x = +a$

Subjective Type Questions :

21. A spring of mass m and stiffness k is fitted to a block of mass M . The block is moving with a constant velocity v on a smooth horizontal surface. The other end of the spring is connected with a rigid wall. Find the maximum compression of the spring before it recoils, assuming that the total energy is conserved.



22. A spring is attached with a block of mass m and a fixed horizontal roof as shown. The block is lying on a smooth horizontal table and initially the spring is vertical and unstretched. Natural length of spring is $3\ell_0$. A constant horizontal force F is applied on the block so that block moves in the direction of force. When length of the spring becomes $5\ell_0$, block leaves contact with the table. Find the constant force F , if initial and final velocity of block is zero.



23. The potential energy of a system of two particles is given by $U(x) = a/x^2 - b/x$. Find the minimum potential energy of the system, where x is the distance of separation and a and b are constants.
24. A body of mass m dropped from a certain height strikes a light vertical fixed spring of stiffness k and sticks to it. Find the height of its fall before touching the spring if the maximum compression of the spring is equal to $\frac{3mg}{k}$.
25. Two blocks of masses m_1 and m_2 are connected by a spring of stiffness k . The coefficient of friction between the blocks and the surface is μ . Find the minimum constant force F to be applied to m_1 in order to just slide the mass m_2 .



ANSWER KEY OF CAPS-05

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|----------------------------------|---|-----------------------------|---|---------|
| 1. (A) | 2. (B) | 3. (D) | 4. (A) | 5. (A) |
| 6. (B) | 7. (C) | 8. (B) | 9. (B) | 10. (C) |
| 11. (BC) | 12. (ABC) | 13. (B) | 14. (C) | 15. (B) |
| 16. (0004) | 17. $\frac{5}{8} \text{ mga}$ | 18. 2 | 19. n^3 | |
| 20. (A) T (B) P (C) Q, R (D) Q | 21. $x = \sqrt{\frac{(3M + m)}{3k}} v.$ | | | |
| 22. $F = \frac{5}{12} \text{mg}$ | 23. $\frac{b^2}{4a}$ | 24. $\frac{3\text{mg}}{2k}$ | 25. $\left[m_1 + \frac{m_2}{2} \right] \mu\text{g}.$ | |